Shaun Eisner

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Research Interests

Ocean-Ecosystem Coupling, Remote Sensing, Bio-optics, Plankton Physiology & Ecology, Ocean Biogeochemistry, Polar Oceanography, Mixed Layer Dynamics, Ocean Data Assimilation

Professional Appointments

Postdoctoral Scholar, Oregon State University — Oregon, USA
Advisor: Michael J. Behrenfeld — March 2026 - Present

Education

Ph.D, Atmospheric and Oceanic Sciences — Maryland, USA
University of Maryland at College Park — 2026
- Advisor: James A. Carton

M.S., Atmospheric and Oceanic Sciences — Maryland, USA
University of Maryland at College Park — 2025

B.S., Physics — Georgia, USA
Georgia Institute of Technology — 2021

Current Research

Modeling Phytoplankton Primary Production
Oregon State University — 2026 - Present

- Using PACE-OCI bio-optical fields (ocean color, particle backscatter) with in situ data and ecosystem models to estimate phytoplankton growth, biomass, and net primary production
- Developing approaches to extend surface satellite signals to subsurface productivity via light–nutrient–grazing-physiology interactions
- Quantifying sub–mixed layer contributions to water-column integrated primary production

Categorizing Phytoplankton Community Responses to Environmental Forcing
Oregon State University — 2026 - Present

- Investigating how satellite-observable phytoplankton properties (ocean color, backscatter, size structure) respond to changes in physical forcing
- Linking variability in stratification, mixed layer depth, and light availability to community composition and primary production across seasonal to interannual timescales
- Identifying common patterns in ecosystem response across biogeographic regions

Impact of Late Summer Storms on Subpolar Phytoplankton Productivity
Dalhousie University & University of Maryland at College Park — 2024 - Present

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- Using a coupled ROMS–sea ice–biogeochemical model to quantify storm-driven nutrient supply and fall phytoplankton production
- Assessing the role of episodic momentum and buoyancy fluxes in modulating seasonal productivity in the Northwest Atlantic

Publications

Journal Articles

1. **Eisner, S. A.**, Carton, J. A., & Chafik, L. (2025). Atlantic water heat transport variability and trends into the Amerasian Basin: A first look using SODA4. *Journal of Geophysical Research: Oceans*. https://doi.org/10.1029/2025JC023382
2. **Eisner, S. A.**, Carton, J. A., Chafik, L., and Smedsrud, L. H. (2026). Increased ocean heat transport to the central Arctic despite a well working Barents Sea Cooling Machine, *Ocean Sci.*, https://doi.org/10.5194/os-22-1073-2026.
3. **Eisner, S. A.**, Chafik, L., Steele, M., & Carton, J. (2026). Long-term Trends and Variability in the Arctic Mixed Layer. *ESS Open Archive*, https://doi.org/10.22541/essoar.177006793.31575234/v1 (Revisions submitted to *JGR: Oceans*)

Book Chapters

1. James Carton, **Shaun Eisner**. *Encyclopedia of Climate System Science*. Chapter 3.1.6: Tropical Atlantic Variability. Elsevier (*in press*).

Selected Presentations

1. **Shaun Eisner**, James Carton. The Impacts of Thermal Fronts and Sea Ice Motion on the Variability of Arctic Near-Surface Currents. *104th Annual AMS Meeting*, 2024 (*Poster*).
2. **Shaun Eisner**, James Carton. Improving Observational Arctic Near-Surface Circulation Products with Constraints from Sea Ice Motion Vectors. *Ocean Sciences Meeting*, 2024. (*Oral*).
3. **Shaun Eisner**, Lee Cooper, Jacqueline Grebmeier, James Carton. Evaluating the Impact of Summer and Fall Storms on Seasonal and Subseasonal Chlorophyll Dynamics in the Bering and Chukchi Seas. *AGU Fall Meeting*, 2024. (*Oral*).
4. **Shaun Eisner**, James Carton, Deirdre Byrne, Semyon Grodsky, Eric Leuliette. The Development of a New Daily Global Mesoscale Blended Ocean Surface Currents (BOSC) Product *EGU General Assembly*, 2023. (*Oral*).

Teaching

Teaching Assistant
University of Maryland at College Park — Fall 2022 - Present

- Course: "AOSC 375 - Introduction to the Blue Ocean". Co-teacher (including lecturing, preparing assignments, grading, and hosting office hours) of an undegraduate course covering introductory topics in oceanography.

Relevant Outreach

Nature Camp Volunteer
Irvine Nature Center — 2022 - Present

- Ongoing volunteer at the Irvine Nature Center teaching earth sciences, environmental sciences, land stewardship, sustainability, and environmental education to children. Activities often involve hands-on experience in natural environments.

Guest Speaker
Baltimore City Public Schools — 2025

- Invited to speak with school-aged children at Connexions Community Arts Schools about ocean sciences and careers in STEM.

Awards and Honors

- **Visiting Scholar - LOREX Program**, Dalhousie University — 2024
- **Undergraduate Research Award**, Georgia Tech College of Sciences — 2020